

INVEST BUDDY

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PROBLEM STATEMENT



Ever thought about investing in the market but felt overwhelmed by too many options and the fear of losing money?

For investors with limited time or market expertise, professionally managed investment options, such as mutual funds and stocks, provide a structured approach where experts handle key decisions. However, navigating the vast array of investment choices can be complex and overwhelming.

80% of investors feel overwhelmed when it comes to investing...



Are you one of them?

~DELOITTE, 2022

if yes...

Meet InvestBuddy – your trusted companion for smarter, stress-free investing!

PROJECT SUMMARY

◆ Retrieve Mutual Fund Data

Fetch key details like fund type, historical performance, fees, and risk metrics.

◆ Search & Filter Mutual Funds

Implement filtering options to refine search results based on user preferences.

◆ Compare Multiple Mutual Funds

Select multiple funds for side-by-side comparison.
Display financial metrics and historical performance trends.

◆ Retrieve Fund Portfolio Details

Extract details about underlying assets in each mutual fund.
Provide insights into sector allocations and major stock holdings.

◆ Visualize Performance Trends

Use data visualization techniques to display fund performance over time.
Generate line charts, bar graphs, and other visual tools to illustrate trends.

◆ Determine Fund Suitability

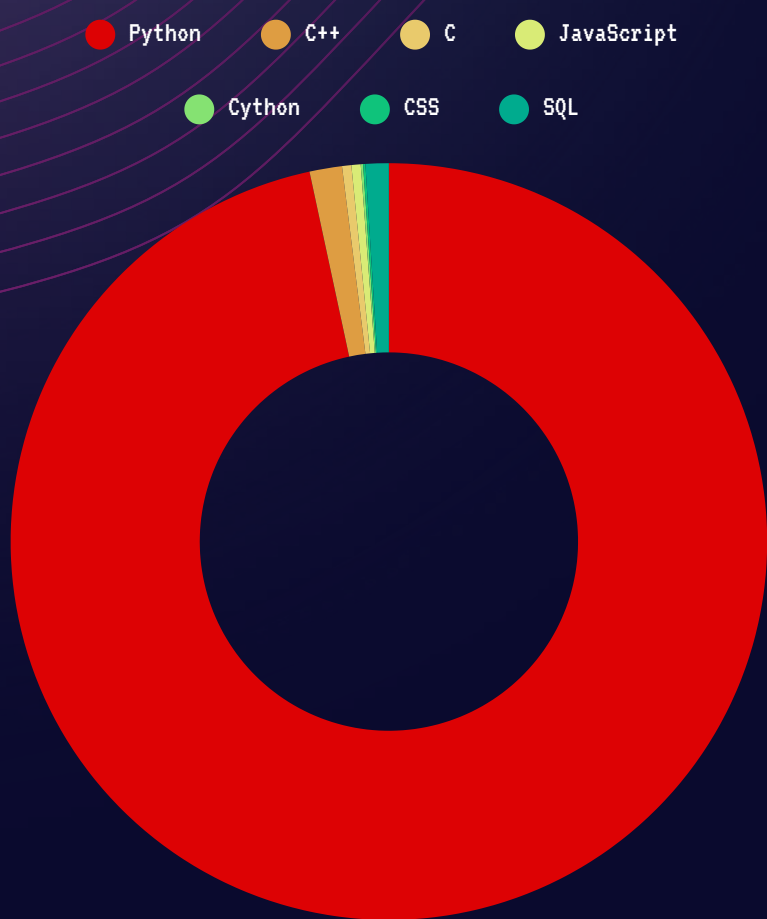
Assess mutual funds based on user-defined preferences (e.g., risk tolerance, investment goals).
Recommend funds that align with the user's profile.

◆ User Profile & Preferences

Allow users to create an account and save investment preferences.
Enable users to track and revisit their preferred funds.



TECH STACKS



Python
C++
C
JavaScript
Cython
CSS
MySQL

REQUIREMENTS

Search & Filter: Easily find and filter mutual funds based on your preferences.

Login Page: Securely access your account and manage investments.

Comparison Feature: Compare multiple mutual funds side by side.

Performance Visualization: Track fund performance over time with interactive charts.

Fund Details: Get in-depth insights into a fund's history and attributes.

Historical Data: Analyze past performance trends for informed decisions.

SIP & Risk Analysis: Evaluate SIP options, risk levels, and fund attributes.

Portfolio Overview: View and manage your investment portfolio effortlessly.

UI SNIPPETS

Our home page provides a list of 814 funds retrieved from kaggle dataset.

Invest Buddy

Home

Compare

Calculate

Personalize

Login

Search Mutual Funds

Select	Scheme Name	Min SIP	Expense Ratio	Fund Size (Cr)	Fund Manager	Category	Alpha	Beta	Sharpe Ratio	Rating	Risk	Returns 1yr	Returns 3yr	Returns 5yr
☐	Aditya Birla SL Active Debt Multi-Mgr FoF-Dir Growth	100.00	0.27	10.00	Kaustubh Gupta	Other	2.24	0.01	0.24	3	9.39	4.0000	6.5000	6.9000
☐	Aditya Birla SL Credit Risk Fund	100.00	0.69	1032.00	Sunaina da Cunha	Debt	3.01	-0.95	1.47	4	2.90	8.7000	9.0000	7.2000
☐	Bank of India Tax Advantage Fund	500.00	1.31	676.00	Alok Singh	Equity	3.48	0.89	1.50	4	16.30	0.9000	29.0000	13.2000
☐	Bank of India Ultra Short Duration Fund	1000.00	0.36	118.00	Mithraem Bharucha	Debt	2.06	1.12	1.07	2	0.52	5.2000	4.7000	5.7000
☐	Baroda BNP Paribas Aggressive Hybrid Fund	300.00	0.61	772.00	Jitendra Sriram	Hybrid	1.88	0.79	1.39	5	11.77	3.5000	20.6000	13.3000
☐	Baroda BNP Paribas Arbitrage Fund	300.00	0.49	354.00	Neeraj Saxena	Hybrid	1.39	0.59	0.85	3	0.71	5.2000	4.6000	5.6000
☐	Baroda BNP Paribas Balanced	500.00	0.64	3123.00	Sanjay Chawla	Hybrid	2.61	0.70	1.46	4	10.57	3.7000	19.8000	0.0000

Upon clicking any fund, the graphs for NAV growth of the fund over time as well as portfolio data is found. We also find all the respective fund details for that fund too and the nav comparison of the selected fund.

Aditya Birla SL Active Debt Multi-Mgr FoF-Dir Growth Allocation

Other Holdings	60.8%
HDFC Bank Ltd.	5.7%
Bajaj Finance Ltd.	5.1%
Cholamandalam Ltd.	4.8%
Divi's Laboratories	3.8%
Dixon Technologies	3.7%
Zomato Ltd.	3.5%
Suven Pharmaceuticals	3.4%
Tata Motors Ltd.	3.3%
NA	2.9%
Cummins India Ltd.	2.8%

Aditya Birla SL Active Debt Multi-Mgr FoF-Dir Growth NAV Over Time

Date	NAV
Tue, 14 Jan 2025 00:00:00 GMT	0.065
Wed, 19 Mar 2025 00:00:00 GMT	0.445

Min SIP

₹100.00

Alpha

2.24

Sharpe Ratio

0.24

Expense Ratio

0.27%

Beta

0.01

Returns (1yr)

4.0000%

Fund Size (Cr)

₹10.00 Cr

Rating

3

Returns (3yr)

6.5000%

Fund Manager

Kaustubh Gupta

Risk

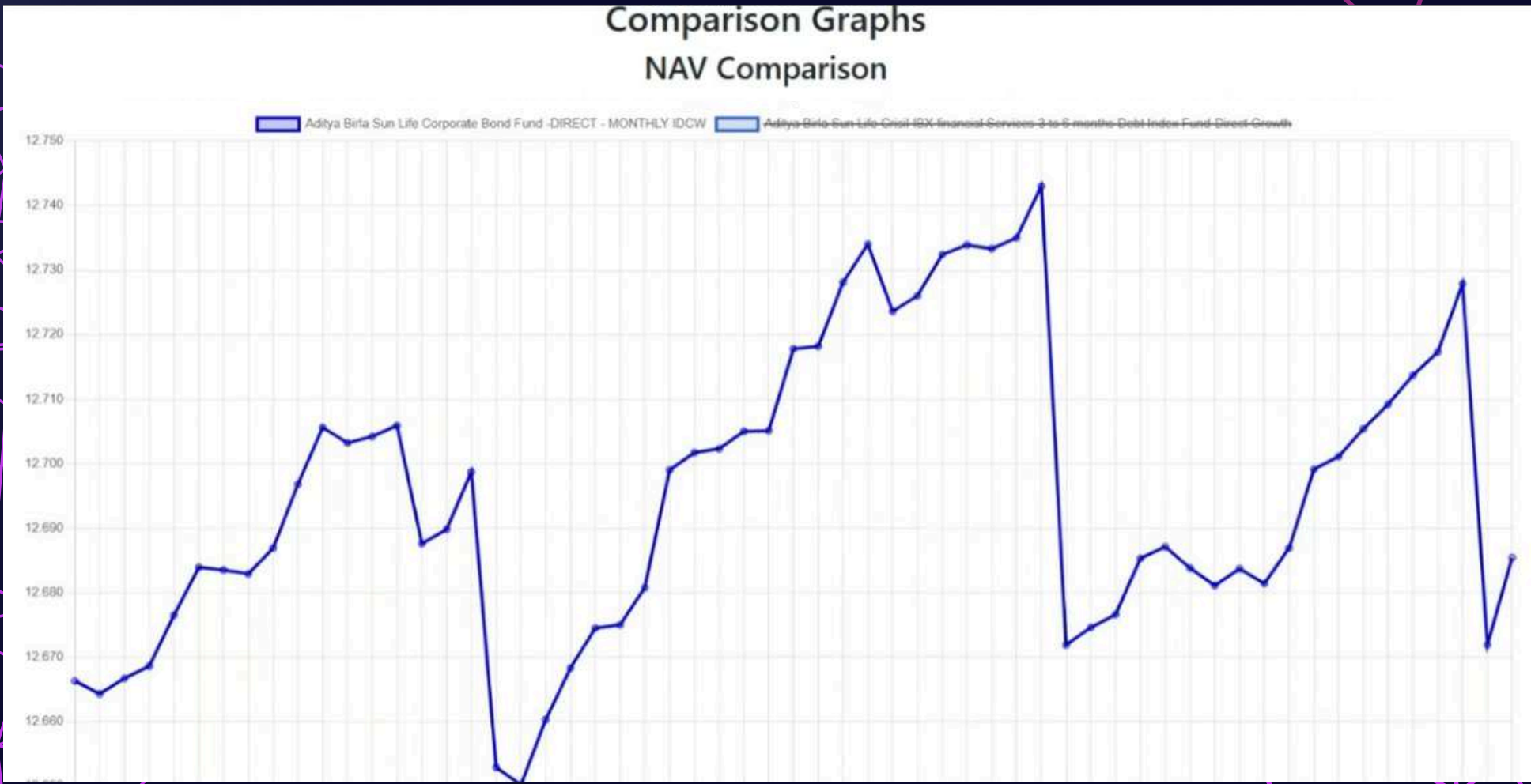
9.39

Returns (5yr)

6.9000%

A search box on the same page to avoid the hassle of manual searching.

Compare Mutual Funds	
360 ONE Focused Equity Fund - Regular Plan - Dividend	
360 ONE LIQUID FUND REGULAR PLAN DAILY DIVIDEND REINVESTMENT	
Aditya Birla Sun Life Balanced Advantage Fund -REGULAR - IDCW	
Aditya Birla Sun Life Corporate Bond Fund -DIRECT - MONTHLY IDCW	
ADITYA BIRLA SUN LIFE CRISIL BROAD BASED GILT ETF	
Aditya Birla Sun Life Crisil IBX 60:40 SDL + AAA PSU APR 2026 Index Fund-Direct Growth	
Aditya Birla Sun Life Crisil IBX 60:40 SDL + AAA PSU APR 2026 Index Fund-Direct IDCW Payout	Compare
Aditya Birla Sun Life Crisil IBX Gilt June 2027 Index Fund-Direct Growth	
Aditya Birla Sun Life Crisil-IBX financial Services 3 to 6 months Debt Index Fund-Direct Growth	
Aditya Birla Sun Life Crisil-IBX Financial Services 3 to 6 Months Debt Index Fund-Regular Growth	
Aditya Birla Sun Life Debt Plus Arbitrage FoF Scheme - DIRECT - IDCW	



Then we have our compare page, which helps to compare multiple funds at the same time.

My Buddy

Home

Compare

Calculate

Personalize

Compare Mutual Funds

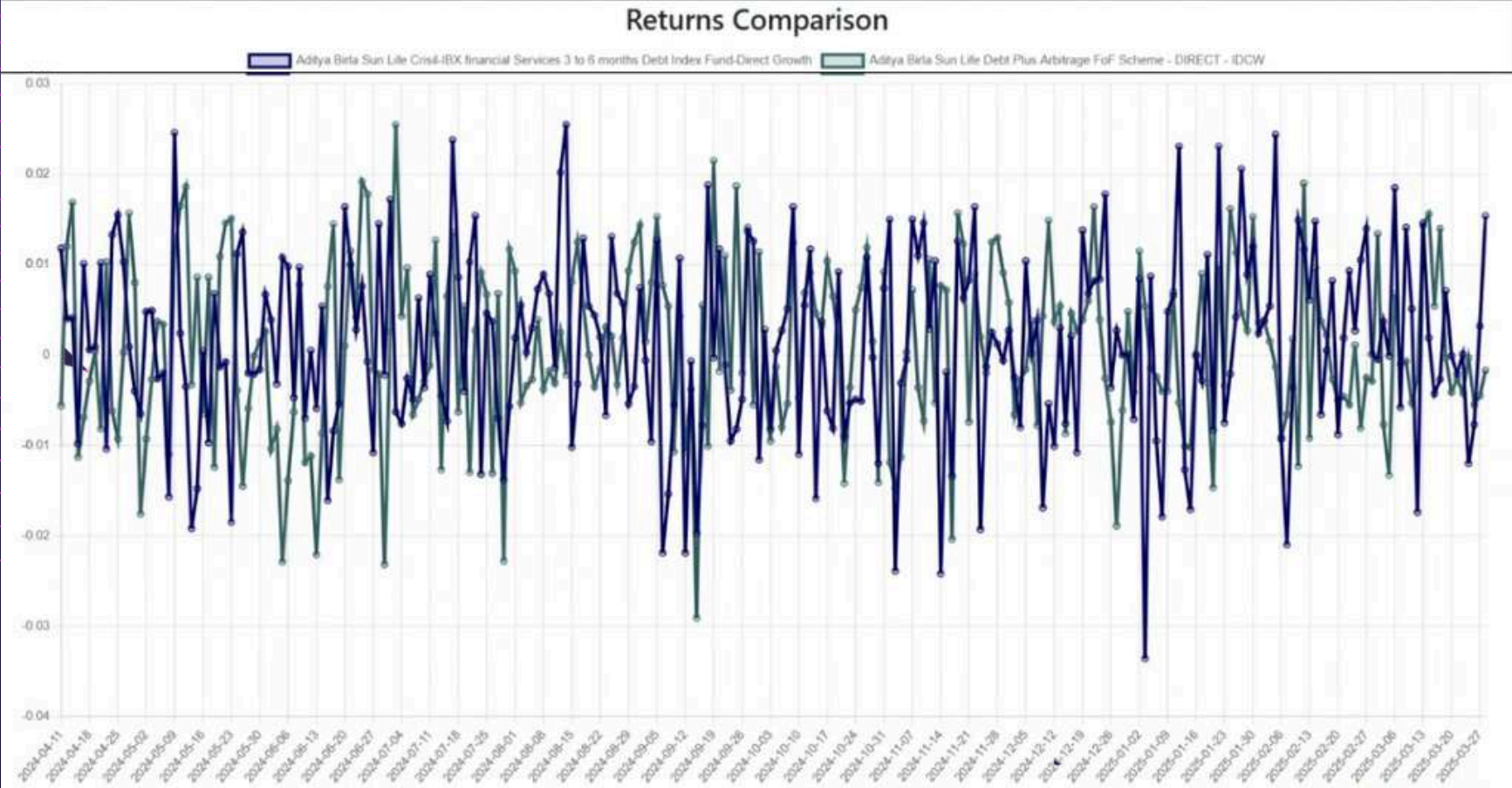
+

+

+

Compare

Upon comparsion, we get graphs for the returns each fund provides.



Upon clicking the calculate page, we are listed with sliders which help filtering on different fund related attributes like returns over different time periods, the fund size, min sip etc.

Invest Buddy

HomeCompareCalculatePersonalizeLogin

Calculate Mutual Funds

Min SIP

0 - 2000

Expense Ratio

0 - 3

Fund Size (Cr)

0 - 60000

Returns 1 Year

-20 - 131

Returns 3 Years

3 - 72

Returns 5 Years

-5 - 33

Category

All

scheme Name	Min SIP	Expense Ratio	Fund Size (Cr)	Fund Manager	Category	Returns 1yr	Returns 3yr	Returns 5yr
Aditya Birla SL Active Debt Multi-Mgr. FoF-Dir Growth	100.00	0.27	10.00	Kaustubh Gupta	Other	4.0000	6.5000	6.9000
Aditya Birla SL Credit Risk Fund	100.00	0.69	1032.00	Sunaina da Cunha	Debt	8.7000	9.0000	7.2000
Bank of India Tax Advantage Fund	500.00	1.31	676.00	Alok Singh	Equity	0.9000	29.0000	13.2000

The category drop down helps to filter further with the type of fund the user wants. For example the user wants Debt category, they can equip it via the drop down

Category

All

Since every user is different, we decided to add a quiz based of which a user can know which fund suits them the best.

Invest Buddy

HomeCompareCalculatePersonalizeLogin

Welcome to the Investment Quiz

This quiz will help us understand your investment preferences and recommend the best mutual funds for you.

Start Quiz

As seen in the snippet below, a series of questions are asked from the user to further know which fund would be the best for them.

Invest Buddy

Home

Compare

Calculate

Personalize

Login

What is your primary investment goal?

☐ Wealth Creation (Long-term growth)

☐ Regular Income (Stable returns)

☐ Tax Saving (Under Section 80C)

☐ Capital Protection (Low-risk investment)

☐ Short-Term Gains (High liquidity needed)

Next

And thus, the final analysis is given, as for what fund would be the best for the individual.

Invest Buddy

Home

Compare

Calculate

Personalize

Login

Recommended Mutual Funds

Axis Bluechip Fund

Category: Wealth Creation (Long-term growth)

Risk: Moderate

Recommended Duration: More than 5 years (Long-term)

Type: Equity Funds (Stock-based, high returns)

And lastly, we have our login page where the user can enter their username, email, password and finally he/she will be logged in.

Invest Buddy

Home

Compare

Calculate

Personalize

Login

Login

Username:

john

Email:

john256@gmail.com

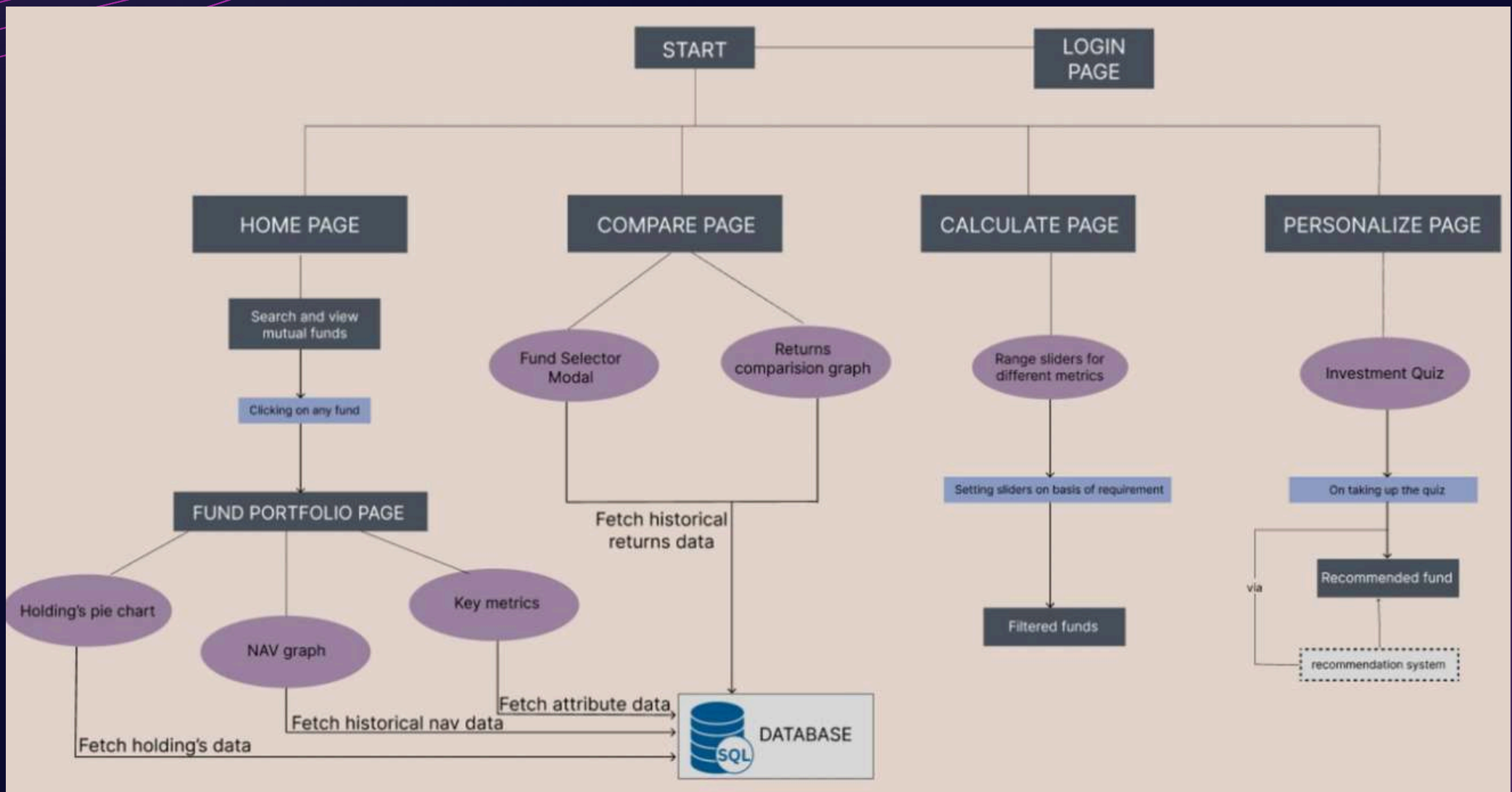
Password:

Login

Welcome, john!

DESIGN AND ARCHITECTURE

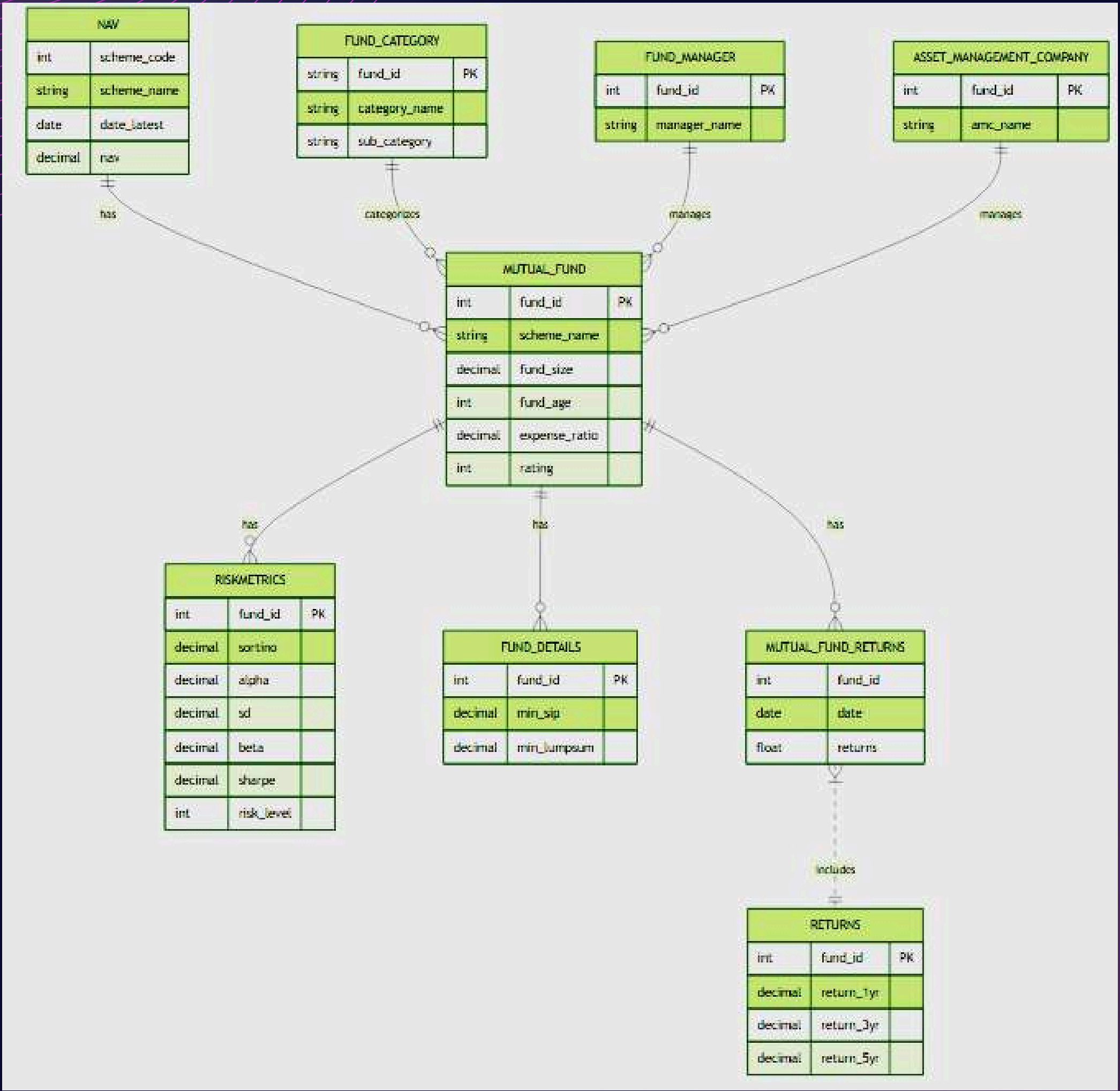
USER FLOW



HIGH-LEVEL DESIGN FOR INVESTBUDDY

- **Architecture:** Web-based application with a frontend (React), backend (Python Flask/Django), and PostgreSQL database.
- **Data Flow:** User requests → Backend processing → Database queries → Response to UI.
- **Modules:**
 - a. User Authentication (Login/Register)
 - b. Mutual Fund Search & View
 - c. Fund Comparison
 - d. Performance Visualization (Graphs, Charts)
 - e. Recommendation System

ER DIAGRAM



ER DIAGRAM FOR MUTUAL FUND MANAGEMENT SYSTEM

THE DIAGRAM REPRESENTS A DATABASE SCHEMA FOR A MUTUAL FUND MANAGEMENT SYSTEM. IT INCLUDES SEVERAL INTERCONNECTED ENTITIES DESIGNED TO TRACK VARIOUS ASPECTS OF MUTUAL FUNDS, SUCH AS THEIR CATEGORIES, MANAGERS, PERFORMANCE METRICS, AND RETURNS.

ENTITIES:

MUTUAL_FUND: CORE ENTITY CONTAINING DETAILS ABOUT MUTUAL FUNDS LIKE SCHEME_NAME, FUND_SIZE, FUND_AGE, EXPENSE_RATIO, AND RATING.

NAV: STORES NET ASSET VALUE (NAV) FOR EACH FUND ALONG WITH SCHEME_CODE AND DATE_LATEST.

FUND_CATEGORY: CATEGORIZES FUNDS BY CATEGORY_NAME AND SUB_CATEGORY.

FUND_MANAGER: MAINTAINS INFORMATION ABOUT FUND MANAGERS (MANAGER_NAME).

ASSET_MANAGEMENT_COMPANY: TRACKS DETAILS OF ASSET MANAGEMENT COMPANIES (AMC_NAME).

RISKMETRICS: CAPTURES VARIOUS RISK METRICS (SORTINO, ALPHA, BETA, SHARPE, ETC.) RELATED TO A FUND.

FUND_DETAILS: SPECIFIES THE MINIMUM SIP AND LUMP-SUM AMOUNTS REQUIRED FOR EACH FUND.

MUTUAL_FUND_RETURNS: RECORDS PERIODIC RETURNS OF A FUND (RETURNS) WITH DATES.

RETURNS: STORES RETURNS OVER DIFFERENT PERIODS (1Y, 3Y, 5Y).

FUTURE SCOPE

- **AI-Powered Chatbot:** A smart assistant integrated with the login system to provide personalized mutual fund recommendations based on user preferences and investment goals.
- **Real-Time Notifications & Alerts:** Instant updates on market trends, fund performance, and personalized investment alerts.
- **Unified Portfolio Tracking:** A comprehensive dashboard linked to user accounts, enabling seamless monitoring of investments across multiple companies.
- **Diversified Investment Insights** – Cover stocks, bonds, commodities, and equities alongside mutual funds.
- **Smart Comparison Tools** – Simplify investment decisions with AI-driven insights and real-time data.
- **Community & Expert Guidance** – Foster discussions, expert insights, and financial education.

REFERENCES

Our dataset can be accessed by clicking [here](#).

KAGGLE DATASET – KAGGLE HOSTS A DIVERSE COLLECTION OF FINANCIAL DATASETS, COVERING MUTUAL FUNDS, STOCK MARKET TRENDS, AND INVESTMENT ANALYTICS.

URL: [KAGGLE DATASET](#)

MFTOOL – MFTOOL IS A PYTHON LIBRARY DESIGNED TO FETCH MUTUAL FUND DATA FROM AMFI (ASSOCIATION OF MUTUAL FUNDS IN INDIA).

URL: [MFTOOL](#)

GROWW – A WIDELY USED INVESTMENT PLATFORM IN INDIA THAT PROVIDES INSIGHTS, TRACKING, AND INVESTMENT OPTIONS FOR STOCKS, MUTUAL FUNDS, AND OTHER FINANCIAL INSTRUMENTS.

URL: [GROWW](#)

ETMONEY – A FINTECH PLATFORM OFFERING MUTUAL FUND INVESTMENTS, EXPENSE TRACKING, AND FINANCIAL PLANNING TOOLS.

URL: [ETMONEY](#)

VALUE RESEARCH – A REPUTED PLATFORM FOR MUTUAL FUND ANALYSIS, OFFERING RESEARCH-BACKED INSIGHTS INTO FUND PERFORMANCE, RATINGS, AND INVESTMENT STRATEGIES.

URL: [VALUE RESEARCH](#)

